EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 16

Assignment: Comprehensive Revision, Integrated Case Study and Examination Preparation

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 16 of 16

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Assignment Instructions

- Answer ALL questions in Sections A, B and C.
- Read the Week 16 module and your lecture notes before attempting each question.
- This 50-mark assignment is revision, not a full end-of-semester examination.
- Draw UML diagrams in correct notation and check that your models are consistent.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define object-oriented analysis and object-oriented design. [3 marks]

Question A2

Distinguish an object from a class. [2 marks]

Question A3

Define encapsulation and abstraction, with one example each. [3 marks]

Question A4

Define inheritance and polymorphism. [2 marks]

Question A5

What is UML? State whether it is a modelling or a programming language. [2 marks]

Question A6

Define an actor and a use case. [2 marks]

Question A7

State the meaning of the multiplicity 1 — 0..*. [2 marks]

Question A8

Differentiate aggregation from composition. [2 marks]

Question A9

Name the four phases of the Unified Process. [2 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

For a mobile money system, users deposit, withdraw, send, receive and check balances; agents serve customers; administrators manage users and monitor transactions. MoneyTransfer is created Pending, completes when the debit and credit commit together, or becomes Failed if validation or persistence fails, leaving balances unchanged. Assume authentication and state any other policy assumptions.

- (a) Identify the actors. [2 marks]
- (b) Identify candidate classes. Propose a justified generalisation, or explain why inheritance should not be forced. [3 marks]
- (c) Draw a sequence diagram for Send Money including failure handling. [4 marks]
- (d) Draw a state-machine diagram for MoneyTransfer using the stated lifecycle. [3 marks]

Question B2 (8 marks)

The sequence diagram calls validateTransaction(), but the corresponding class contains only checkTransaction().

- (a) Identify the modelling problem. [2 marks]
- (b) Explain why consistency matters between diagrams. [3 marks]
- (c) Describe how the team should resolve the discrepancy. [3 marks]

Section C — Case Study (10 marks)

A university wants online student registration. Students log in, view courses, check prerequisites, register, drop courses and view registrations. Administrators create or modify courses, close registration and view reports. Course starts Created, becomes Available when opened with capacity, becomes Full when its last seat is reserved, returns to Available when a seat is released while registration remains open, and becomes Closed when registration is closed. No new registration is allowed in Closed. Each Registration links one Student and one Course, and only one active registration per student/course pair is permitted.

- (a) Identify the actors and use cases. [3 marks]
- (b) Draw a class diagram with relationships and multiplicities. [4 marks]
- (c) Draw a state-machine diagram for Course with event labels and guards. [3 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (9 questions)	20
B	Application questions (12 + 8)	20
C	Online student registration	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This is individual work. Plagiarism or copying attracts a penalty under university regulations.
- Use correct UML notation, including multiplicities, messages, events and guards.
- Cite sources using the format in the Week 16 module references.